



Local Artificial Intelligence running on Docker and a Web User Interface

Purpose

This guide/analysis is meant for personnel who wish to set up an Open Source AI that is locally run on the personnel's network. This setup also includes the creation of a User Interface to access the local AI through Python or Docker. The Open Source AI allows any data that the personnel enters to stay private instead of under the property of the company that created the AI.

Lab Commands

ollama run <AI model>

This will both run the AI of choice and download if it is not installed already. Make sure that you specify the parameter count.

docker run

This creates a fresh docker container. You can only use this command once for the same container then use the docker exec command to enter configuration.

docker exec

This enters the docker container that was already created. This command will be used after the docker run command.

DVTM (Dynamic Virtual Terminal Manager)

A function that you can be installed on Linux that allows users to run multiple terminals simultaneously within one terminal window by splitting the window into multiple sections.

Curl - <command>

The curl command is used to download and upload data on Linux

Background

The 2 main applications that are used to set up a Local Ai on any personal laptops are Ollama and LM studio. In this analysis/guide, we set up Ollama over LM studio due to the fact that LM studio has a more simple user interface (UI) which is easy for any personnel to navigate while LM studio has a more customized but more complex user interface. Ollama also has the benefit of having a UI that is compatible with Linux.

Ollama is an open source framework that is used as a lightweight, command line interface(CLI) tool for running large language(LLMs) locally on your personal computer regardless of your computer's operating system. This tool besides providing easy ai model management also is a simple application programming interface (API) to integrate local AI into other applications. The commands for downloading, running and removing these AIs are done in the command line. In order to run these AI models, your computer needs to meet stringent hardware requirements to endure the heavy vram usage. Also in addition, when selecting an open source model to locally host on your computer the model must be of a smaller framework then your computer can handle in order to better handle more complex

and elaborate questions that require an in-depth response without heavily straining the computer's hardware.

However, no matter how developed your computer that you wish to locally host an AI is, it will not be able to match the AIs of large corporations. The reason that huge corporations like OpenAI will have better AIs in terms of speed and reasoning than locally hosted AIs is due to the fact that these companies have financial resources in the billions to spend on large data centers that are able to handle a larger data load, top researchers who can further the development of the AI, and can have very efficient data management.

One of the key benefits of locally hosting AIs than using the models of these large corporations is privacy. Any information whether sensitive or confidential inputted into these AIs made by these corporations will be tracked, used to train other models which can leak your information, and will be sold to data brokers for profit. However, local AIs allow any entity to keep the data inputted within their own organization greatly minimizing the chance that employees will be kept private and not sold.

This concern has greatly increase over the past few years as the interest in AIs have exploded due to the fact that there have been many incidents and will continue to be where employees have consulted AIs such as ChatGPT, Gemini, and CoPilot about sensitive corporate documents which have been used to train these models which answered other personnel who have inquired about the topic with the direct confidential information in response. Specifically, there is a large concern for the use of AI by teachers in educational institutions. Using public AIs can violate FERPA by improperly sharing student personally identifiable information with a third party, as AI tools will use the data without adequate privacy safeguards. This falls under an unauthorized disclosure of student education records, which FERPA protects.

Focusing back on Ollama, the biggest disadvantage is the fact that it does not natively support a user interface on windows, which is the operating system that most personnel use. To setup a user interface on your computer for Ollama, the most optimal choice is to use Open WebUI and put it in a docker container to run in Linux. Although using a Virtual Machine is a way to locally host an AI, the most optimal choice for us it use a program called Docker.

Docker is a core essential software that is being used in this analysis/guide. Docker is a group of platform as a service (PaaS) product that uses OS-level virtualization to deliver software in packages called containers. The service has both free and premium tiers with the free tier being the software that is used in this lab. The software that hosts these containers is called Docker Engine, released in 2013 and developed by Docker. Inc.

Docker essentially is a tool that is used to automate the deployment of applications in lightweight containers so that the applications can work efficiently in different environments. The reason that Docker uses containers is because they can be isolated from another and communicate with each other through well-defined channels. Due to the fact that all the containers share the services of a single operating system kernel, there are fewer resources used than virtual machines. The container can run in a variety of locations such as

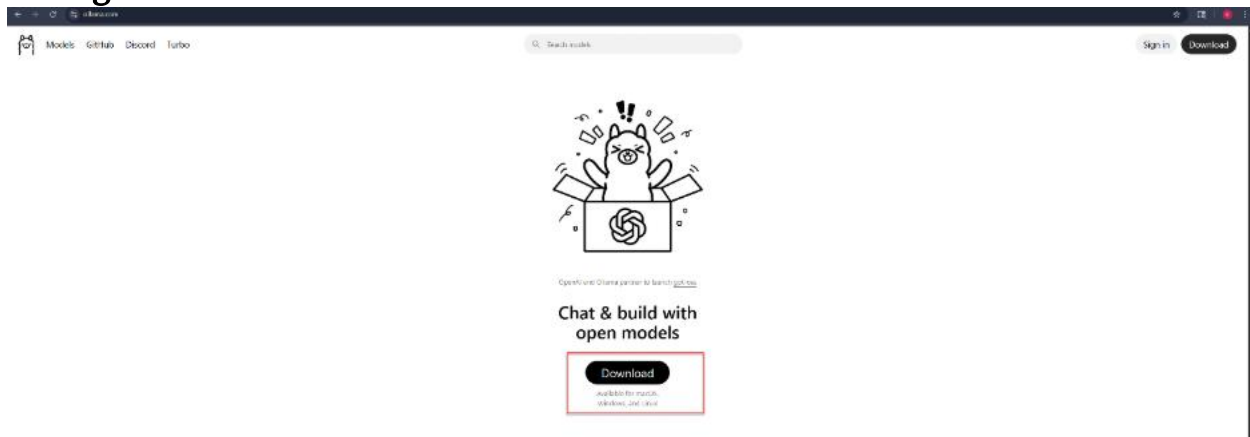
on-premises, in public or private clouds. Since Docker containers are lightweight, a single server of virtual machine can run several containers simultaneously and can even be installed on a single board computer like the Raspberry Pi.

There are 3 components of the Docker software as a service: Software, Objects and Registries. For Software, the Docker Daemon is a process that manages the Docker containers, handles container objects and listens for requests that are sent via the Docker Engine API. The Docker client program provides a CLI that allows users to interact with Docker daemons.

For Objects, Docker Objects are various entities used to assemble an application in Docker. The main classes of Docker objects are images, containers, and services. A container is a standardized, encapsulated environment that runs applications and managed using the Docker API or CLI. An image is a read-only template used to build containers and used to store and ship applications. A service allows containers to be scaled across multiple Docker daemons which is called a swarm, a set of cooperating daemons that communicate through the Docker API.

For Docker Registries, a Registry is a repository for Docker images that clients connect to, in order to download or upload these images that they have built. Registries can be public or private, with the main public registry being Docker Hub. Docker Hub is the default registry where Docker looks for images and allows the creation of notifications based on events.

Configurations



Go to following websites: <https://ollama.com/> and click the download button as highlighted in the image above.

Download Ollama



macOS



Linux

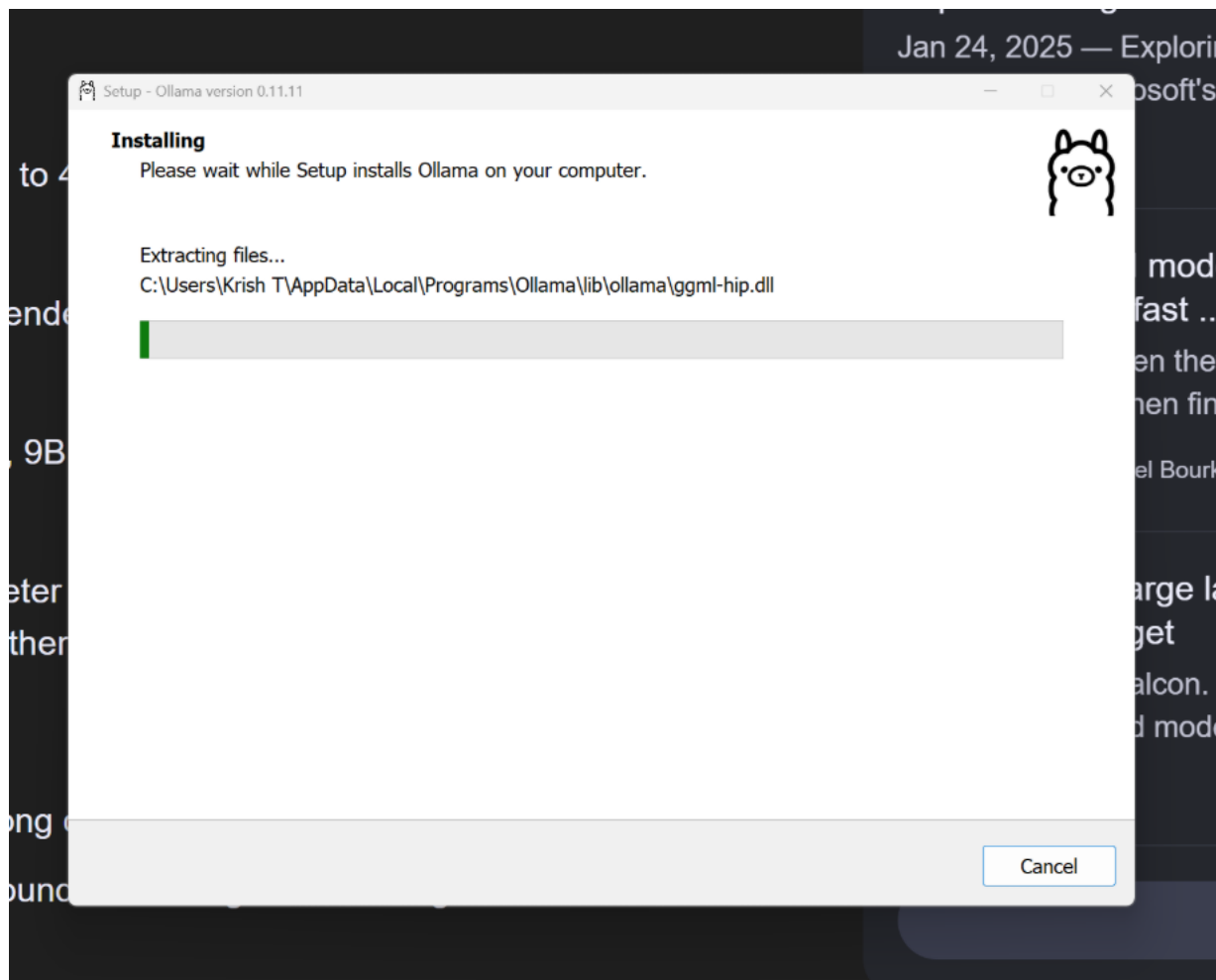


Windows

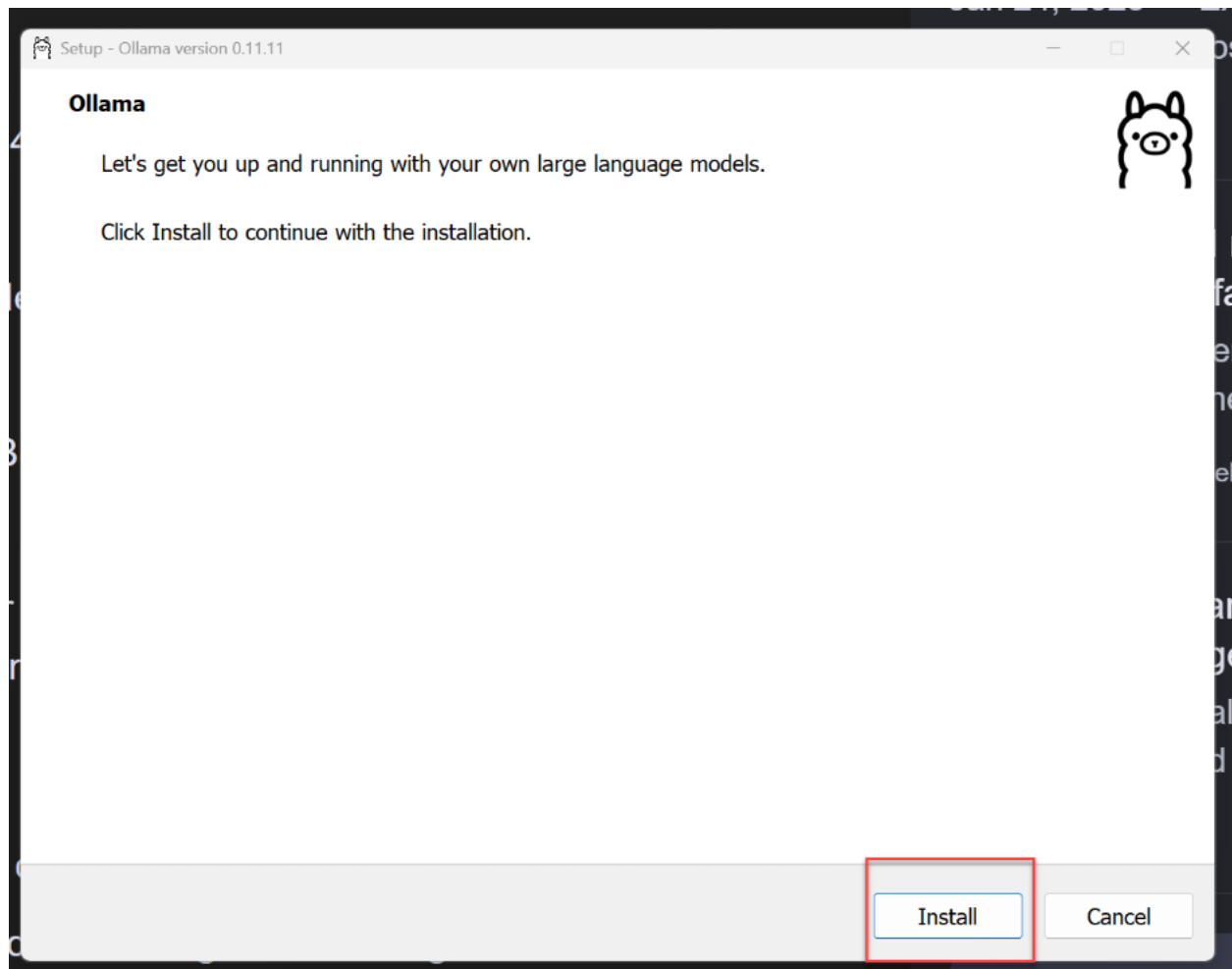
Download for Windows

Requires Windows 10 or later

Select the operating system that your device is running on and then click the Download button as highlighted in the image above.

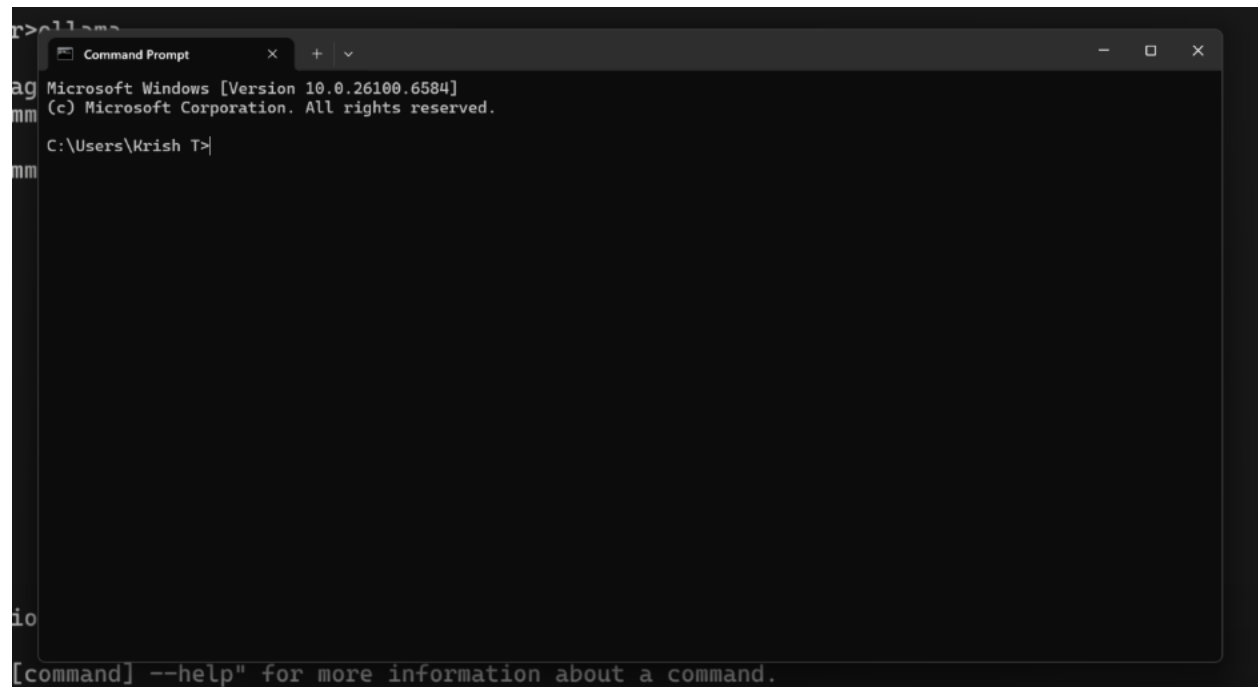


The Download page should then open up a window similar to the image above.



Select the Install button as highlighted in the image above.

```
r>llama
ag
mm
mm
io
[command] --help" for more information about a command.
```



Open the command prompt on your device which should look like the image above.


```
Command Prompt
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Krish T>ollama
Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve    Start ollama
  create   Create a model
  show     Show information for a model
  run      Run a model
  stop     Stop a running model
  pull     Pull a model from a registry
  push     Push a model to a registry
  list     List models
  ps       List running models
  cp       Copy a model
  rm       Remove a model
  help     Help about any command

Flags:
  -h, --help      help for ollama
  -v, --version    Show version information

Use "ollama [command] --help" for more information about a command.

C:\Users\Krish T>
```

Type in “*ollama*” as highlighted in the image above in order to verify that ollama has been downloaded on your device.

```
Command Prompt - ollama
(c) Microsoft Corporation. All rights reserved.

C:\Users\Krish T>ollama
Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve    Start ollama
  create   Create a model
  show     Show information for a model
  run      Run a model
  stop     Stop a running model
  pull     Pull a model from a registry
  push     Push a model to a registry
  list     List models
  ps       List running models
  cp       Copy a model
  rm       Remove a model
  help     Help about any command

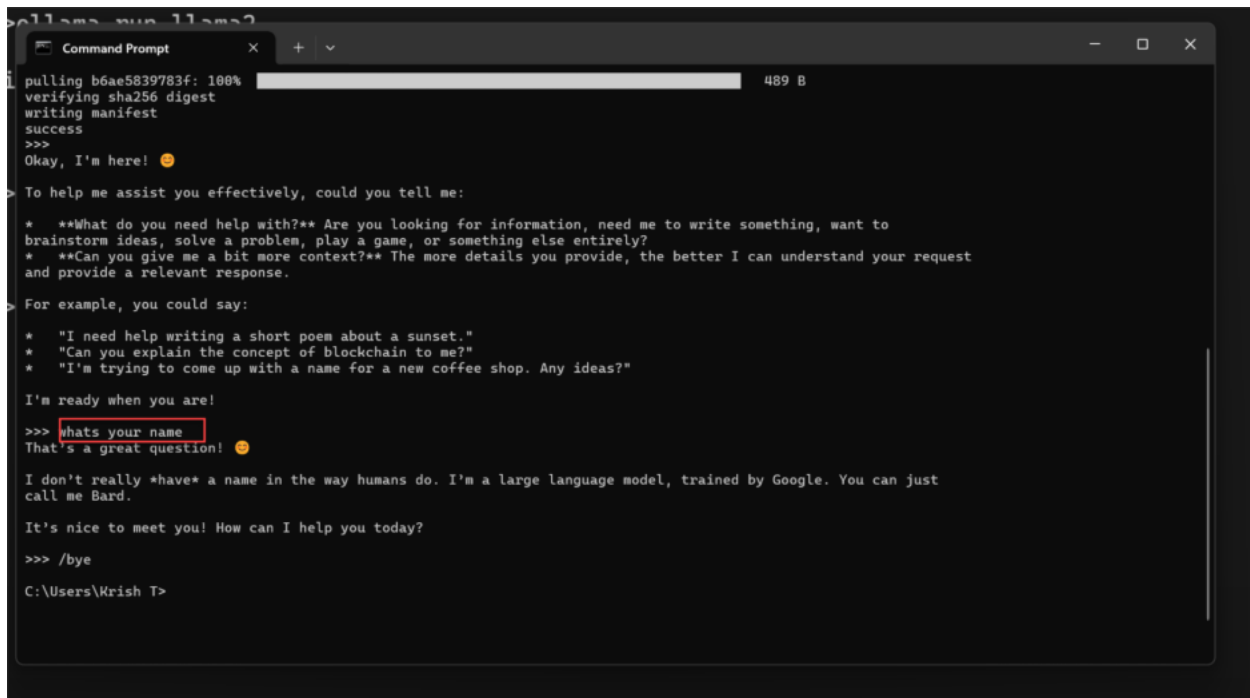
Flags:
  -h, --help      help for ollama
  -v, --version    Show version information

Use "ollama [command] --help" for more information about a command.

C:\Users\Krish T>ollama run gemma3
pulling manifest
pulling aeda25e63ebd: 1% | 25 MB/3.3 GB 4.2
```

Type in “*ollama run*” followed by the specific name of the local ai model that you wish to download with the specific parameters that are specified on the website:

<https://ollama.com/search>. The website shows all the various open source models that the user can use as their local open-source model.



```
ollama pull llama3
Command Prompt
pulling b6ae5839783f: 100% 489 B
verifying sha256 digest
writing manifest
success
>>>
Okay, I'm here! 🤖

> To help me assist you effectively, could you tell me:

* **What do you need help with?** Are you looking for information, need me to write something, want to brainstorm ideas, solve a problem, play a game, or something else entirely?
* **Can you give me a bit more context?** The more details you provide, the better I can understand your request and provide a relevant response.

> For example, you could say:

* "I need help writing a short poem about a sunset."
* "Can you explain the concept of blockchain to me?"
* "I'm trying to come up with a name for a new coffee shop. Any ideas?"

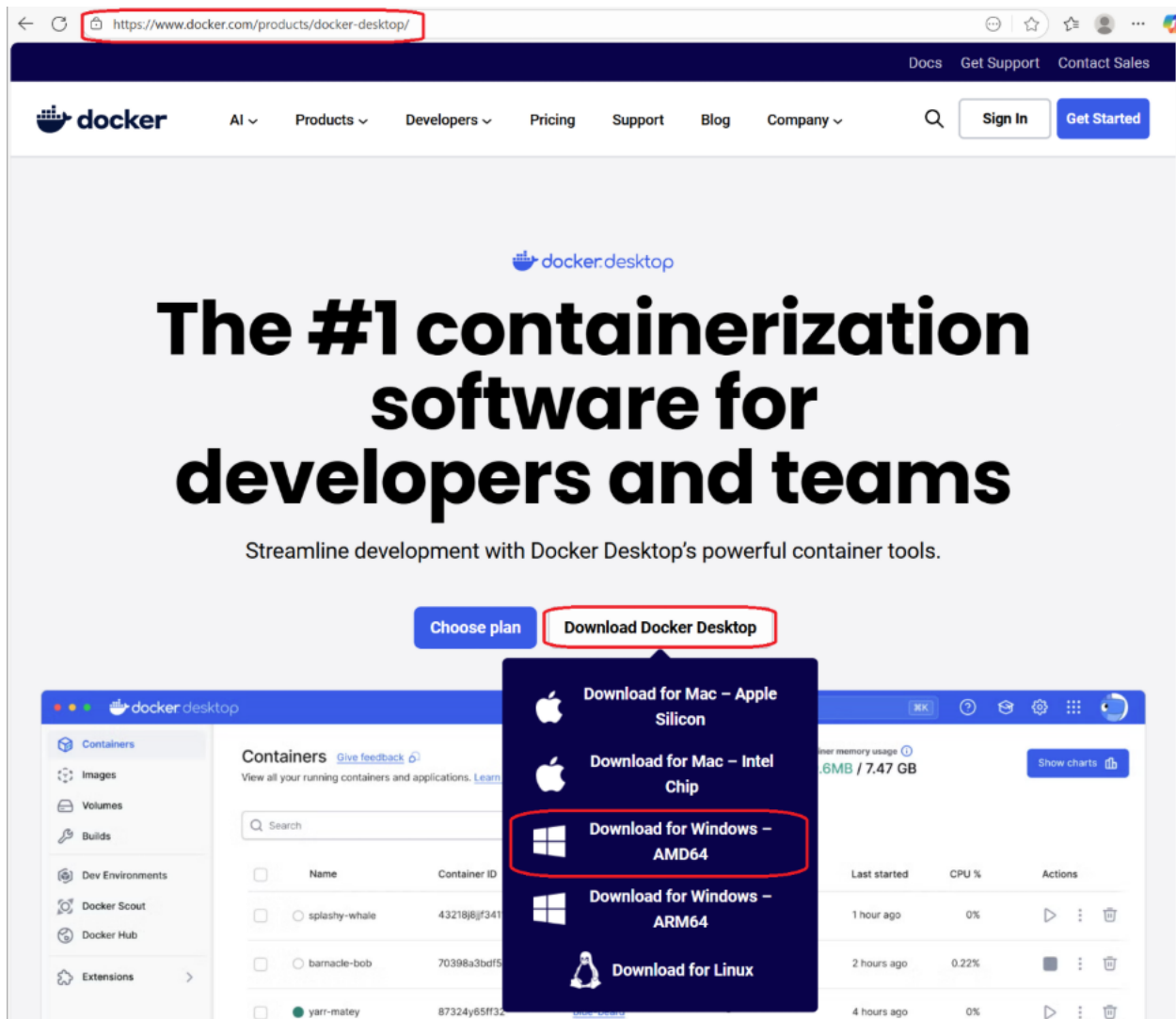
I'm ready when you are!
>>> whats your name
That's a great question! 🤖

I don't really *have* a name in the way humans do. I'm a large language model, trained by Google. You can just call me Bard.

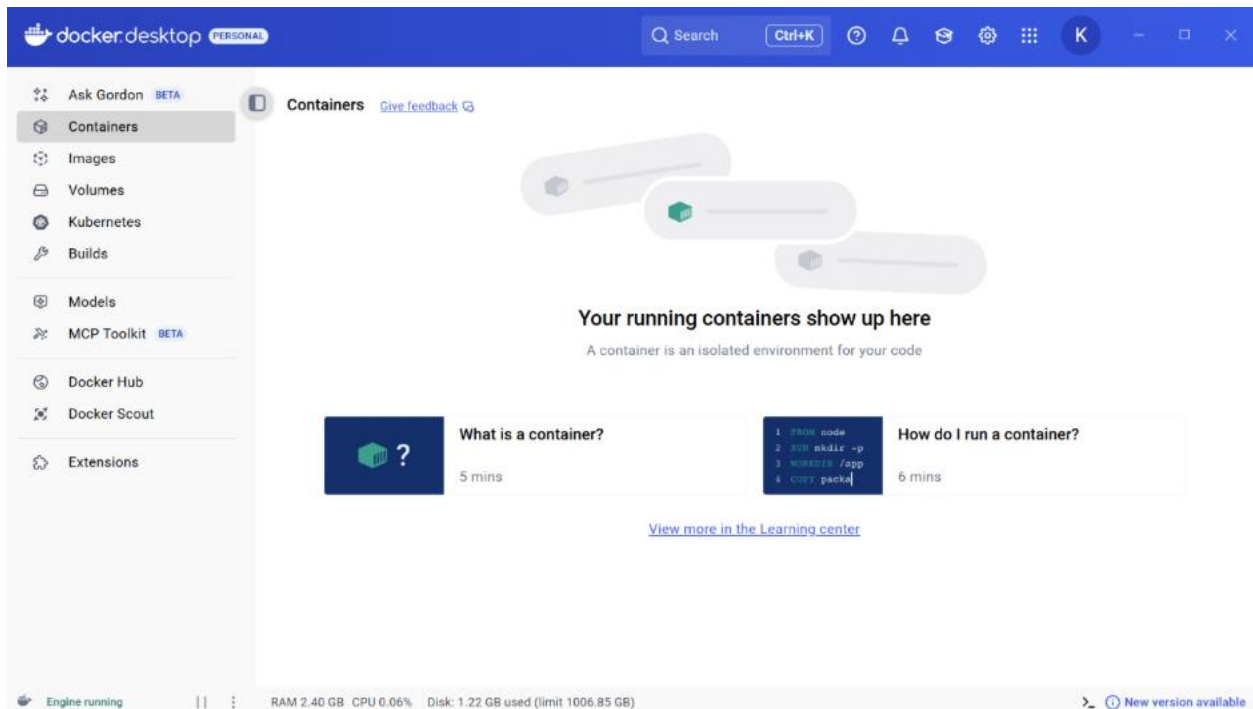
It's nice to meet you! How can I help you today?

>>> /bye
C:\Users\Krish T>
```

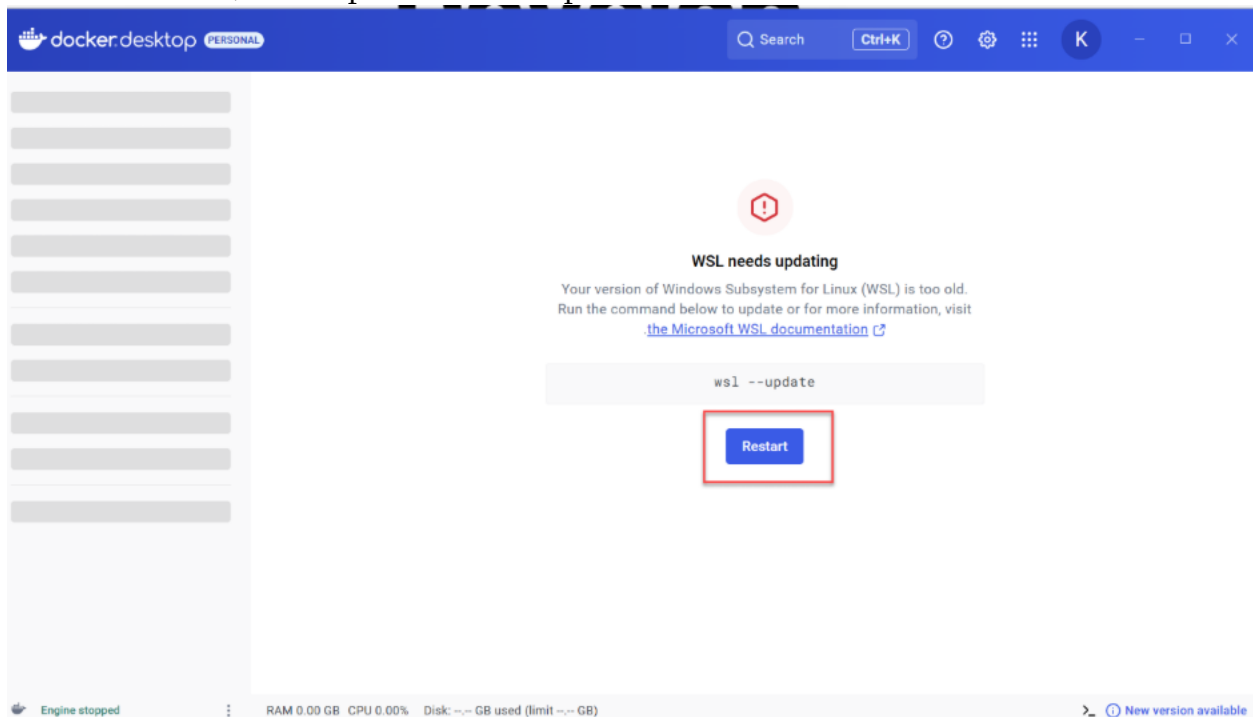
The model will be able to use immediately after the download is complete as highlighted in the image above.



In order to set up the local AI to work on a Web User Interface we will use Docker and Open WebUI. First go to <https://www.docker.com/products/docker-desktop/> and click the Download button for your device's operating system as highlighted in the image.



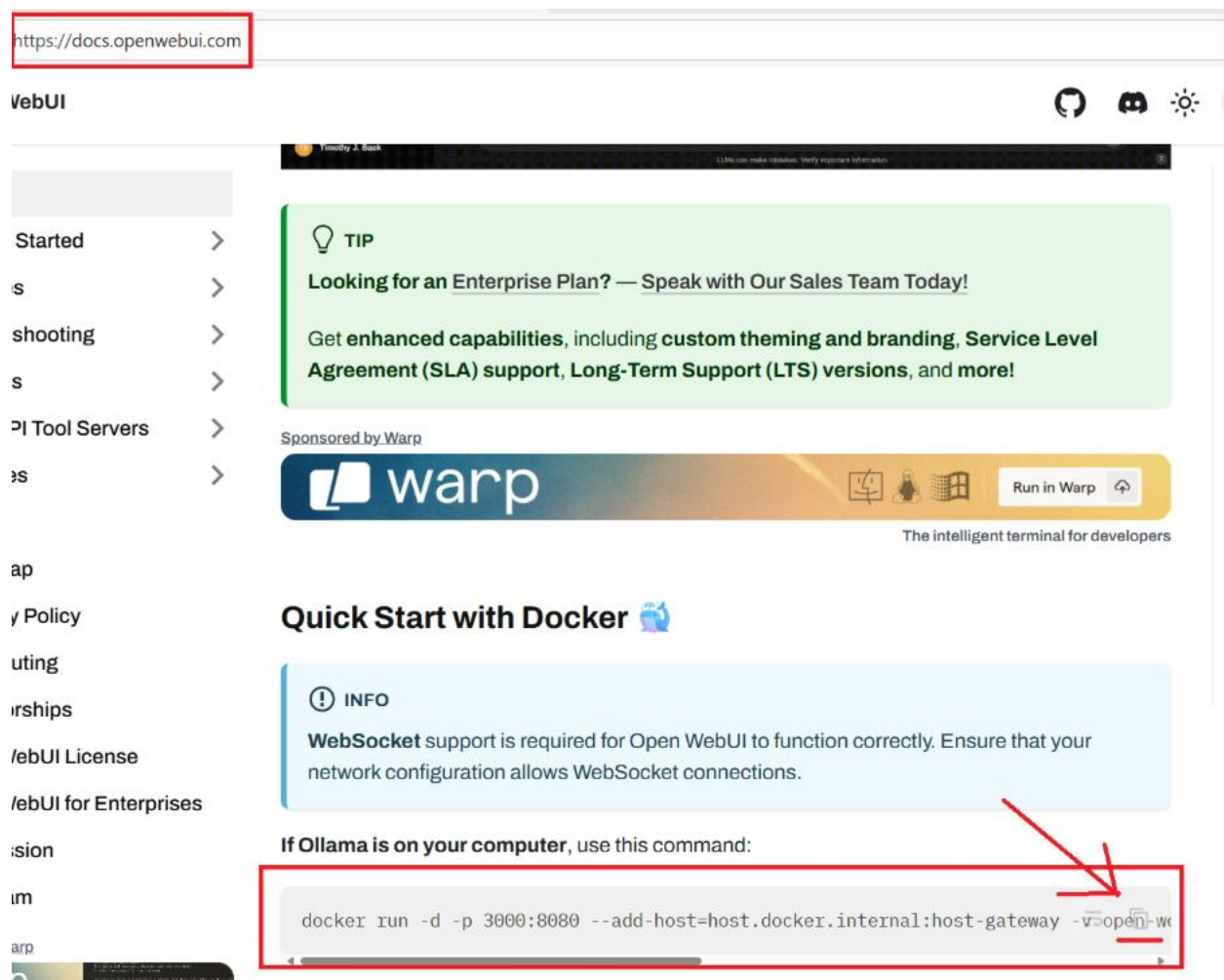
After the installation is complete, open the docker desktop app. Then create a Docker Account. Click to the containers tab on the left and side of the GUI. There should not be any active containers, but keep this window open in order to later create a container.



If a notification as seen in the image above appears click the restart button as highlighted in the image above. This will restart your device to ensure that everything is properly saved.

```
PS C:\Users\Luke M> docker --version
Docker version 28.4.0, build d8eb465
```

Next open the PowerShell application on your device and type the command “docker --version” as highlighted in the image above. This verifies that docker is installed on your device.



Click the following link: <https://docs.openwebui.com> and scroll down to "If Ollama is on your computer" section. Then click the copy button of the first docker run command under the text.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

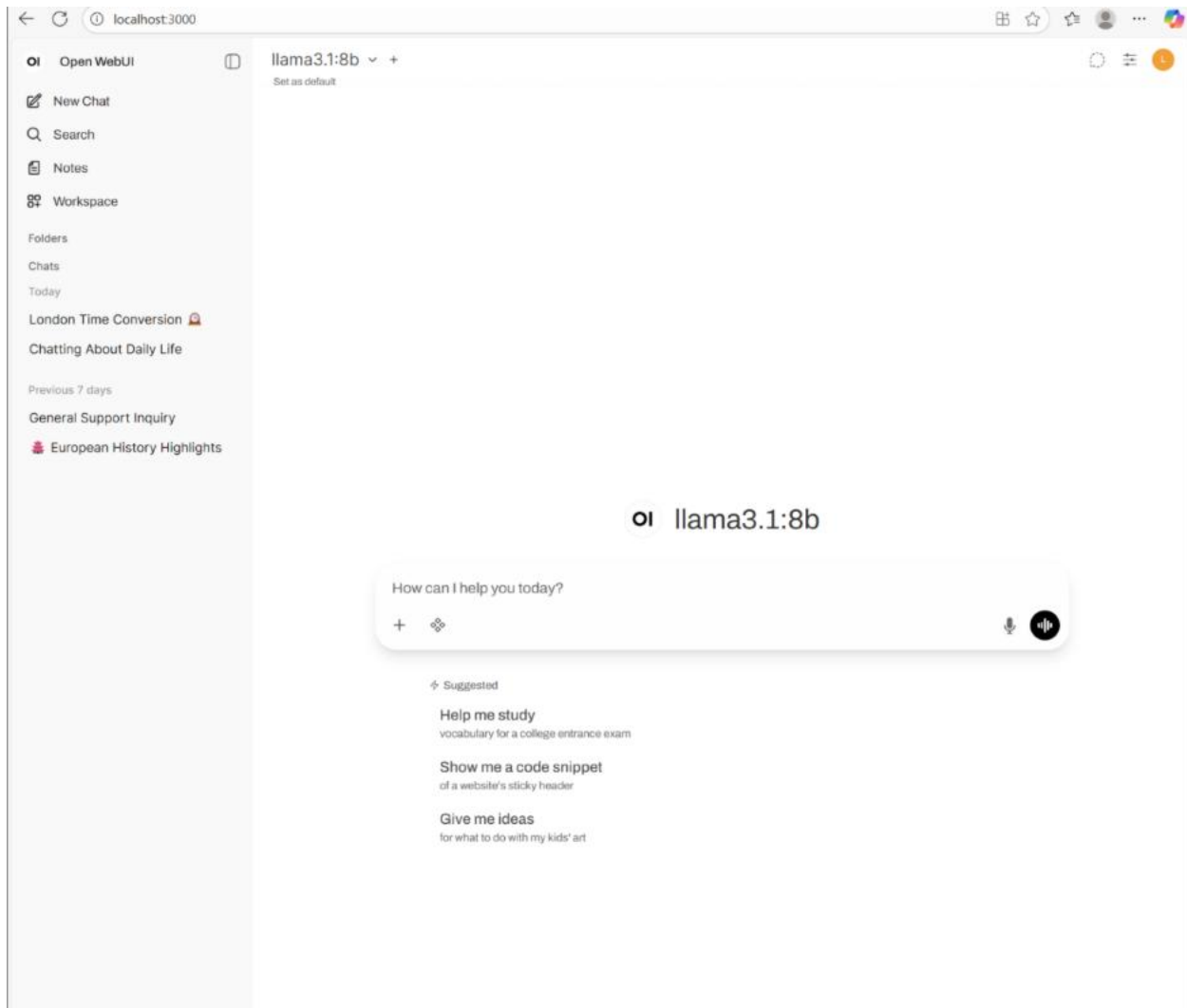
PS C:\Users\Luke M> docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway -v open-webui:/app/backend/data --name open-webui --restart always ghcr.io/open-webui/open-webui:main
```

Next reopen PowerShell and right click within the window in order to paste the command

that was just copied into PowerShell. This will create a container in docker which isolates the apps inside the container from the rest of your device's operating system. A minimal form of Linux should be already installed in the container before the docker run command was issued.

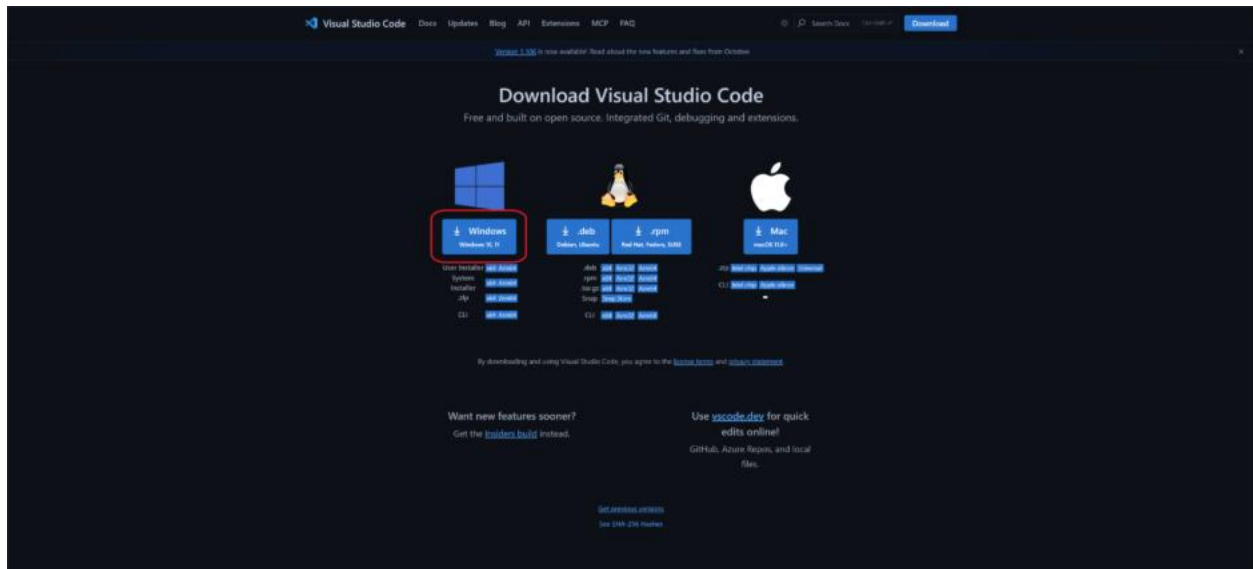


Next open up a browser of your choice and in the search bar type “localhost:3000” This will bring you to a webpage that is created by Open WebUI. Then, create an account and login into that account as shown in the image above.

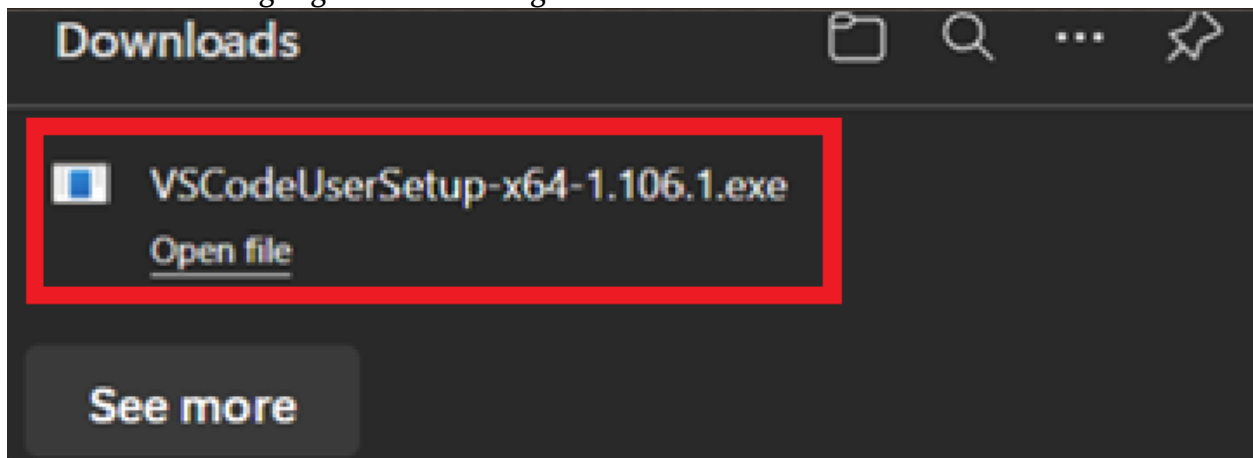


Now you are able to completely interact with the local AI that just got installed. In order to locally connect to the AI from another device, open command prompt and enter the command “*ipconfig*”. Then from the other device that you wish to connect to the local AI open a browser of your choice and in the URL section type your “IP address:3000”. For example, this could look like “*192.168.3.1:3000*” Next, you will be able to use the local AI that is hosted on your first device. Also, throughout this process make sure that both devices are on the same network.

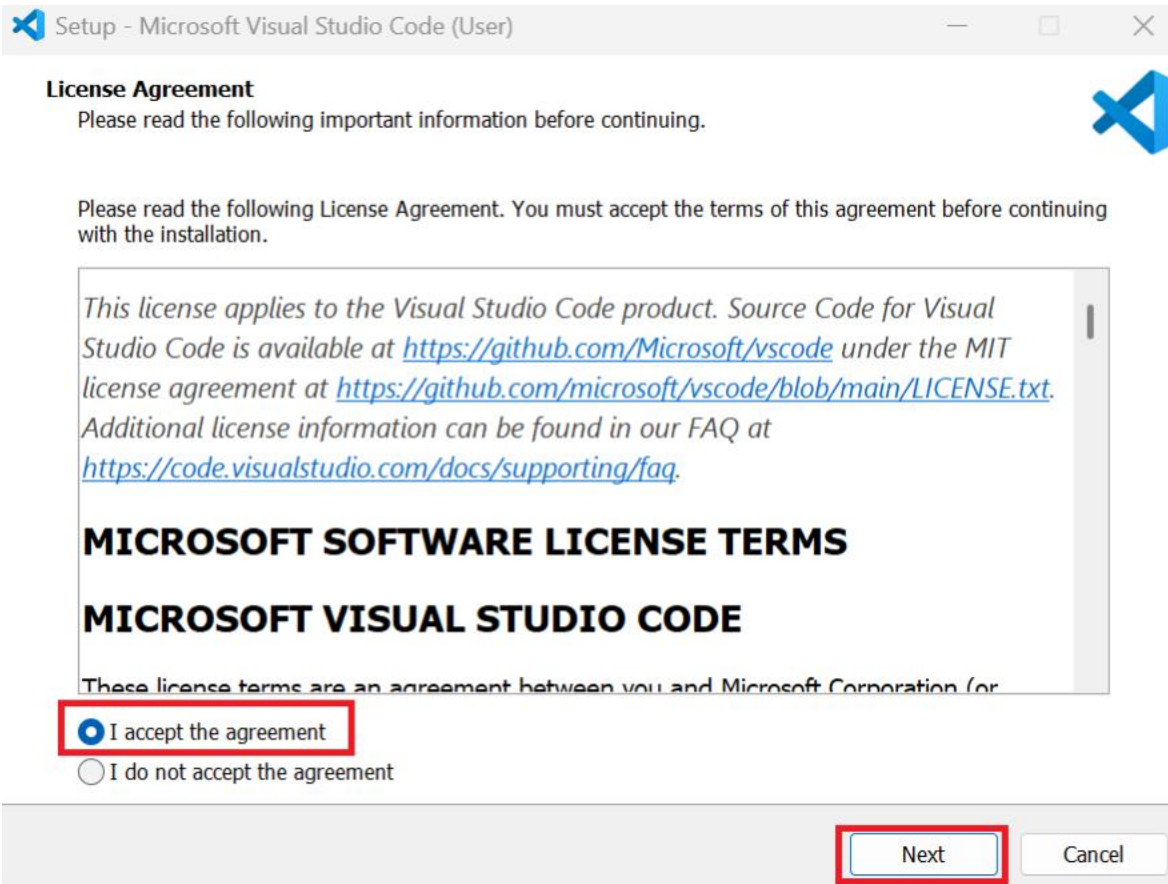
If you don't want to run the local AI on Open WebUI there is another way to run the local AI on your browser. This method relies on using a python script along with html to make the UI and run it locally. To run the Python script, you first need to install VS code. Click the link: <https://code.visualstudio.com/download>.



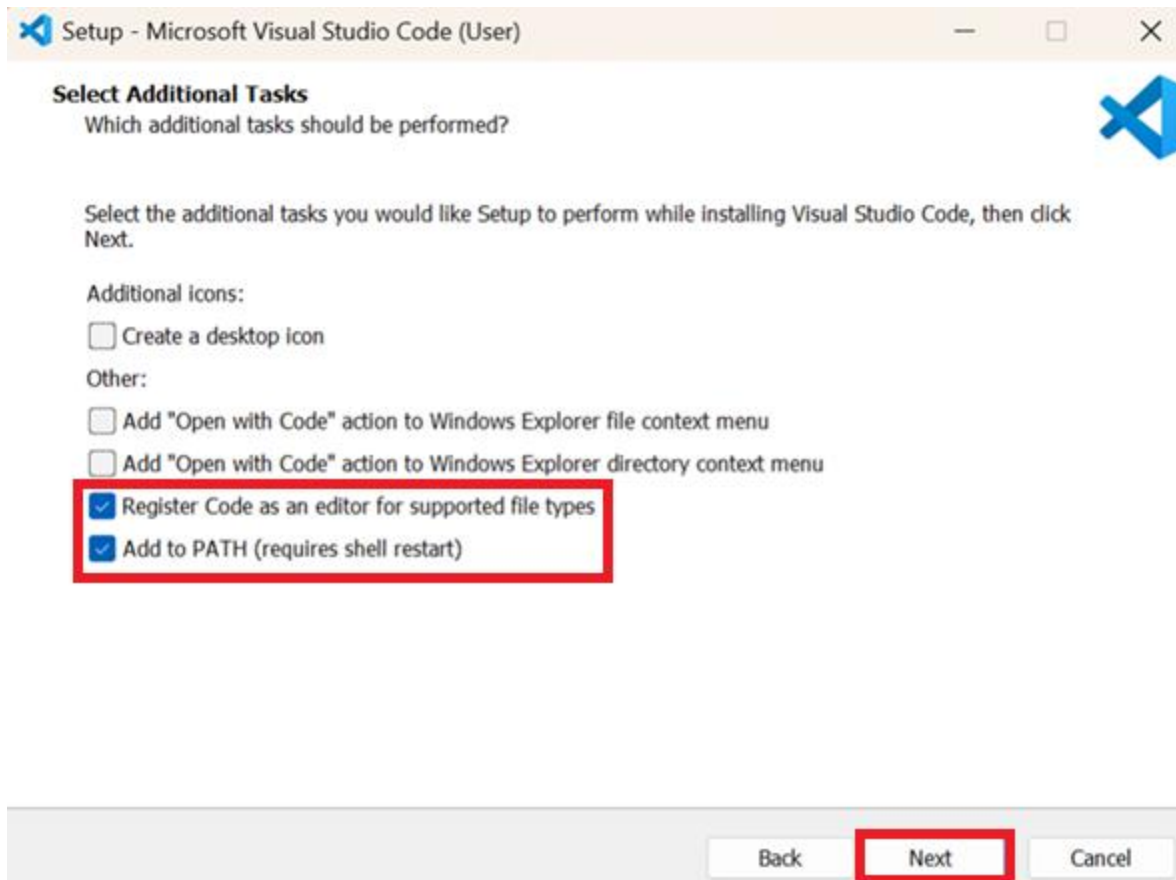
Click on the Operating System that your device is using and press the install button which is demonstrated as highlighted in the image above.



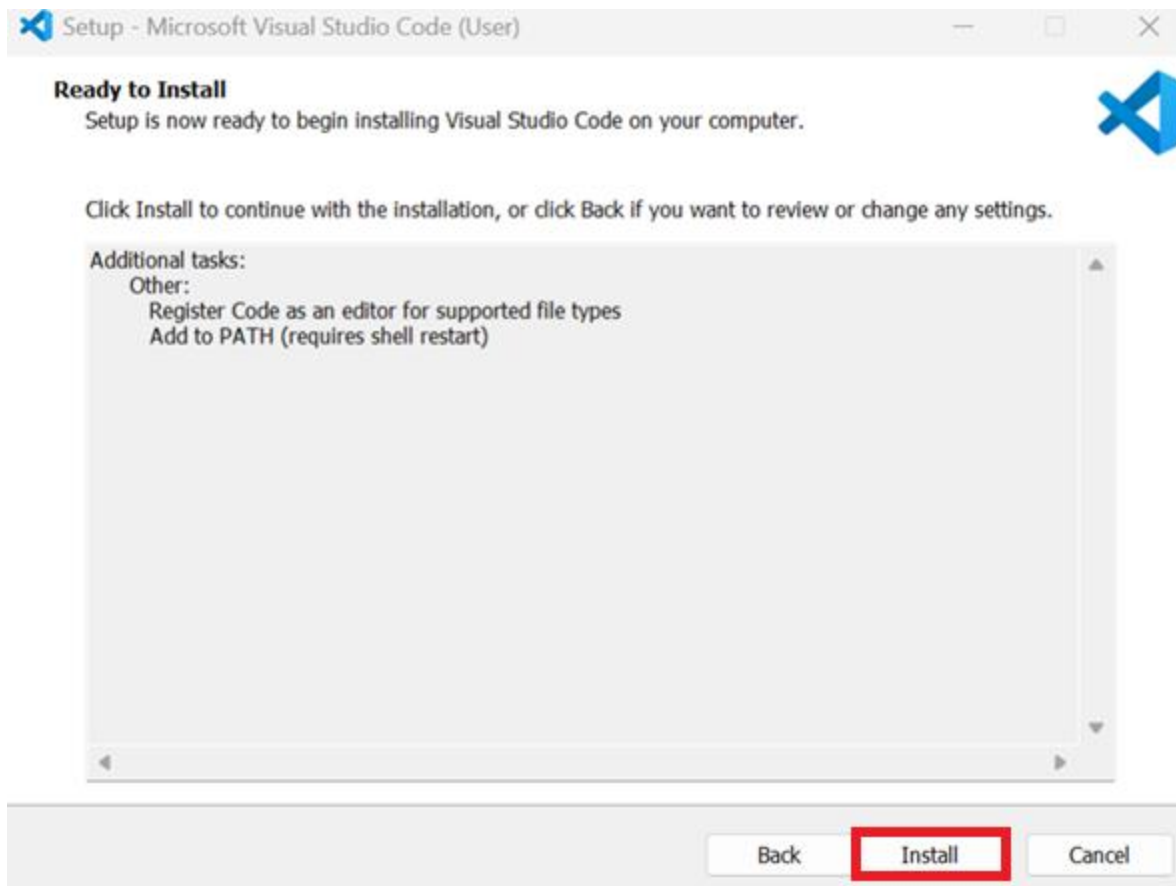
In the top right corner of your browser click the downloaded vs code file as highlighted in the image above.



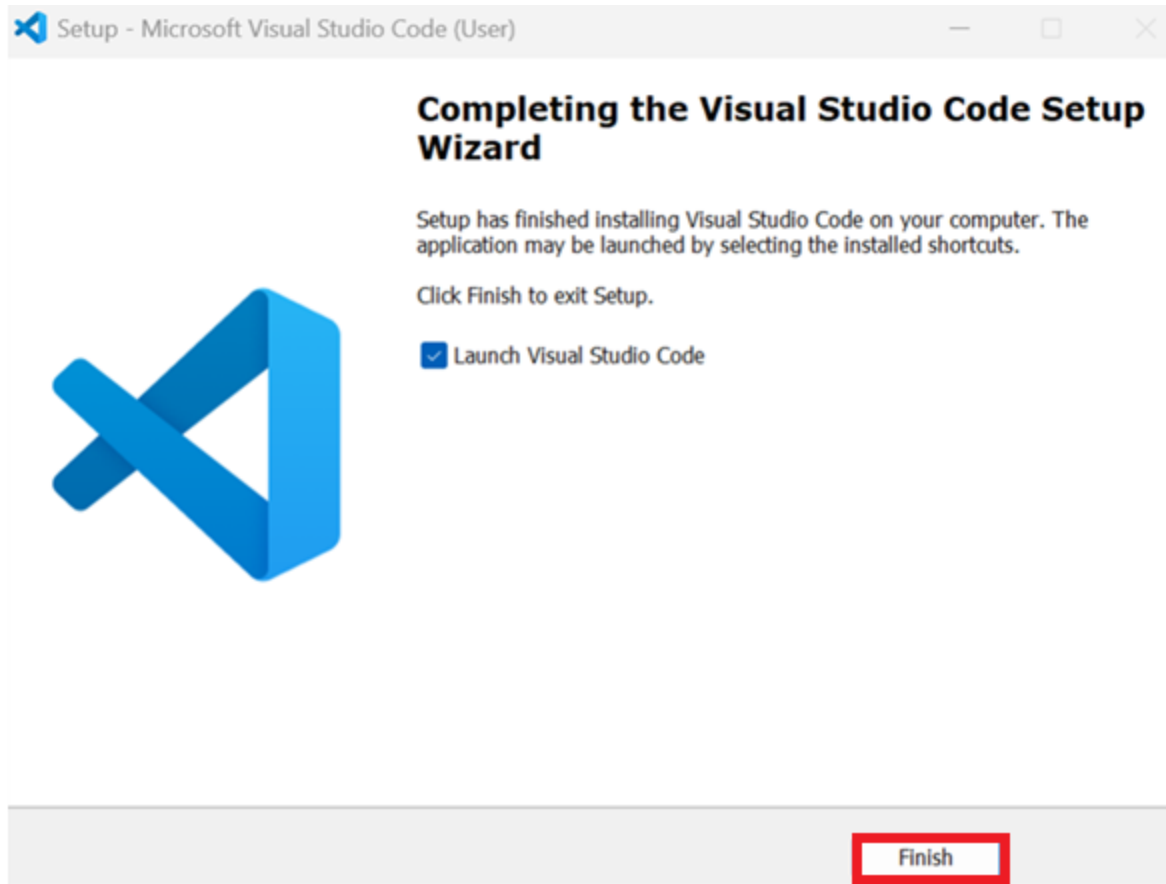
Then, there will be a VS Code Setup that pops up, be sure to check the Agree button and click the Next button as highlighted in the image above.



Check the boxes that highlighted in the image above as well as any other options that you would like to add. Then click the Next button as highlighted in the image above.

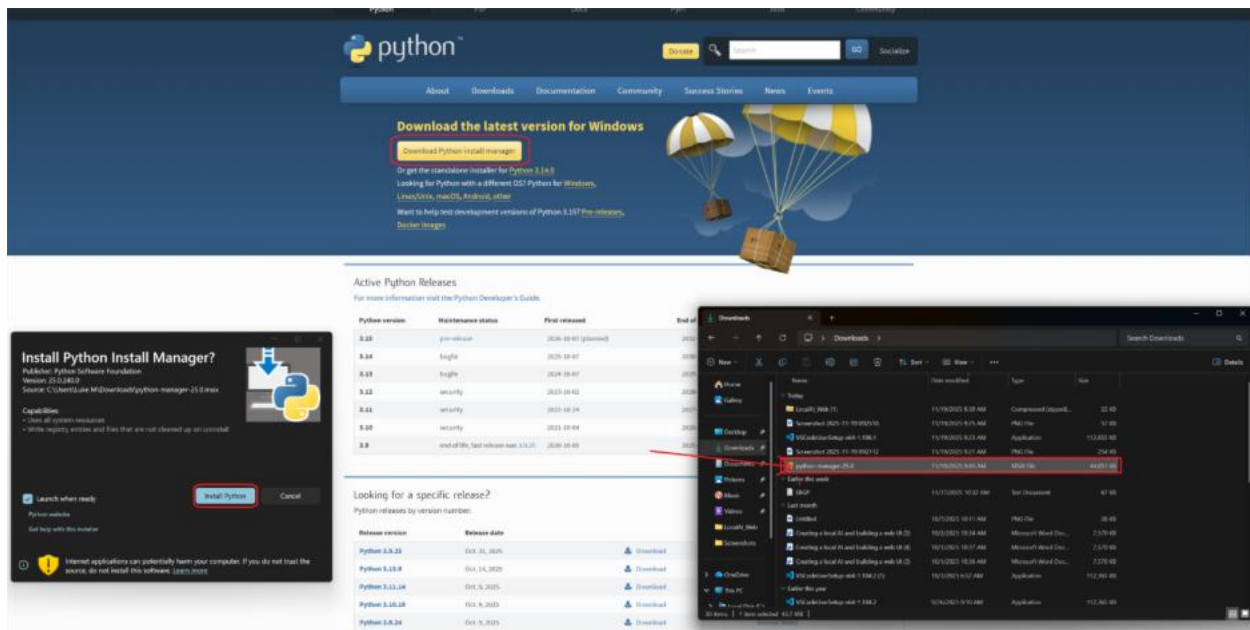


Next select the Install button as highlighted in the image above.



Then click the Finish button as highlighted in the image above.

Now we must install python. Click the following link: <https://www.python.org/downloads/>



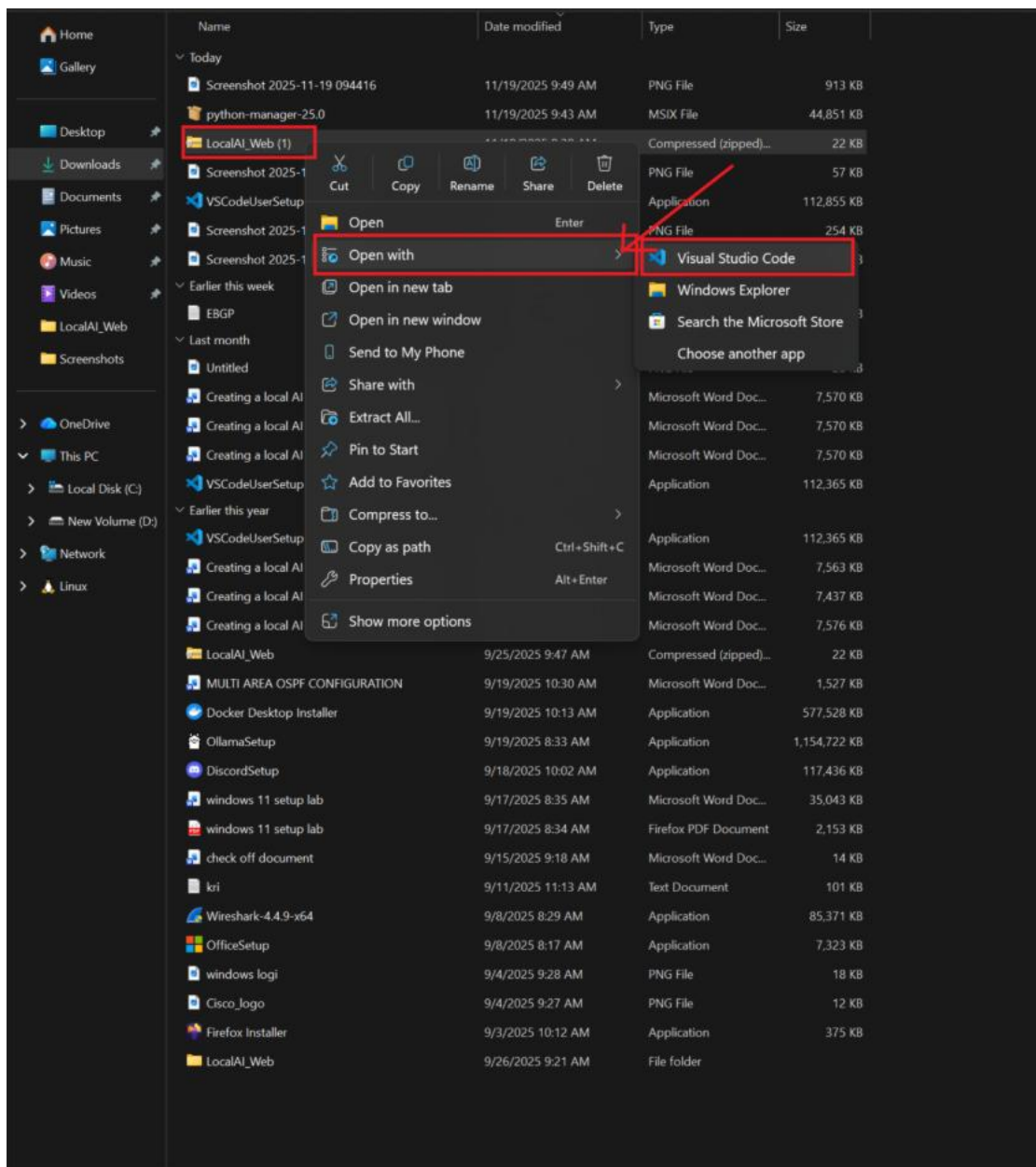
The next step is to download python on your device. Click on the following link:

<https://www.python.org/downloads/>

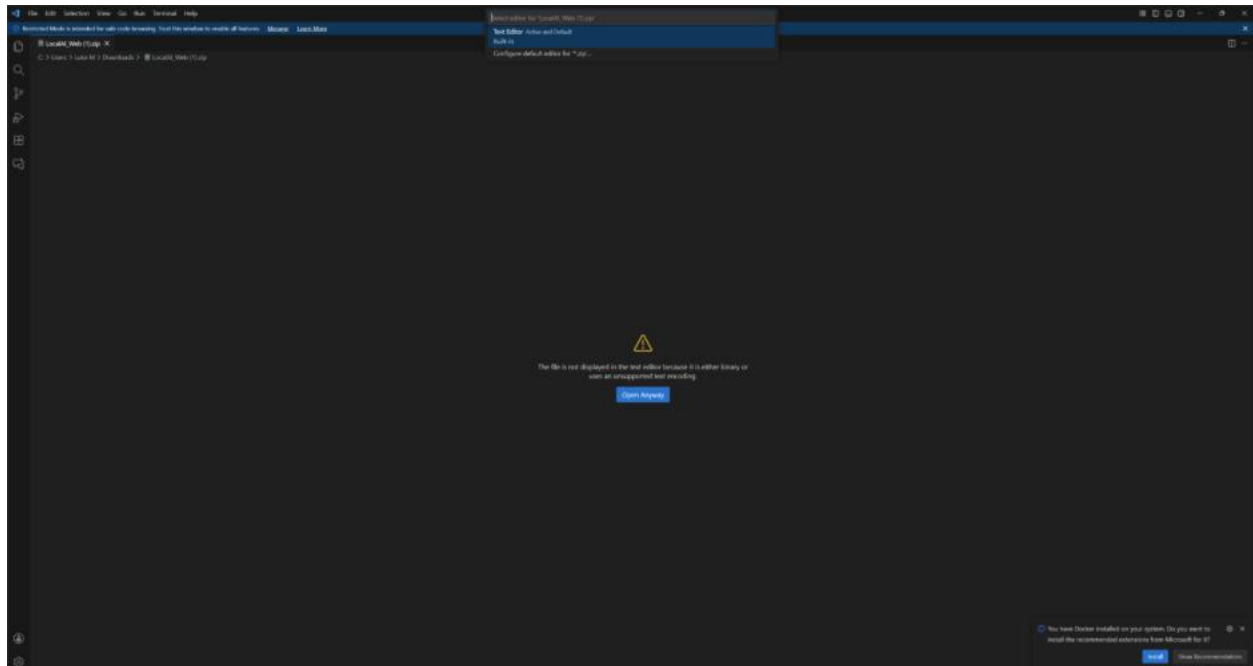
Download the install manager and then open the app installer and click install python

If you want to use the code that we made install this: <https://bsd405-my.sharepoint.com/my?id=%2Fpersonal%2Fs%2Dthapaliyak%5Fbsd405%5Forg%2FDocuments%2FLocalAI%5FWeb%2Ezip&parent=%2Fpersonal%2Fs%2Dthapaliyak%5Fbsd405%5Forg%2FDocuments&ga=1>

Open the file of the link that you clicked above.

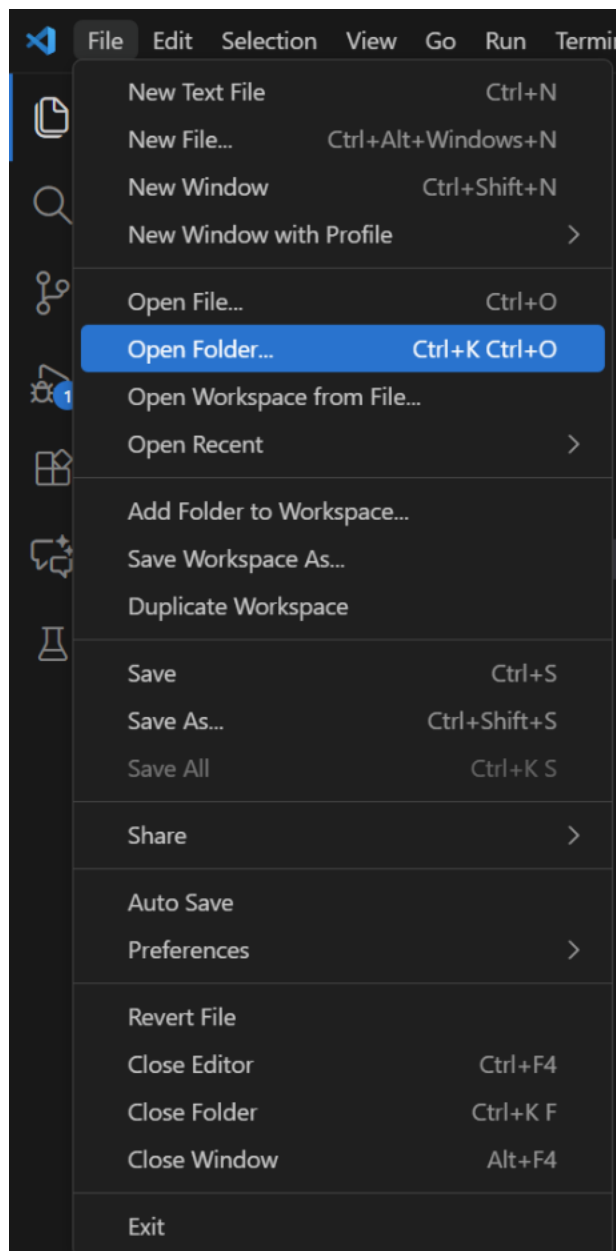


Right click on the zip file and choose open with VS Code or choose the Choose another app and search for VS Code if it is not in the list.

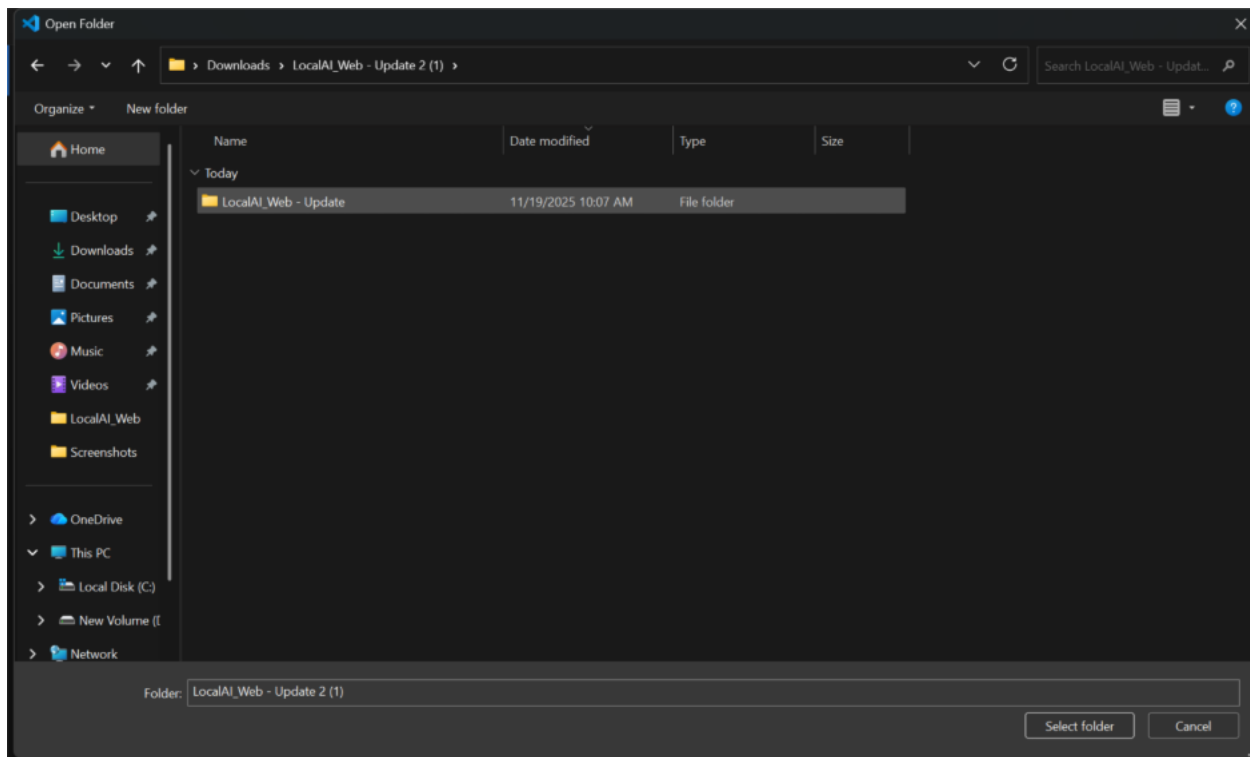


After opening the file select open anyway and then at the top of the VS code window choose the text editor.

If the file does not work, find the script and extract it. Then, close the script in VS Code.



Click the file button in the top left corner of the window and then the Open folder button.



Find the file in your Onedrive and double click on the file which should look like this. Then, select the file and click select folder.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\server.py", line 10, in <module>
    from analysis import analyze_prompt
File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\analysis.py", line 5, in <module>
    from web_scraper import fetch_html
File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\web_scraper.py", line 1, in <module>
    import requests
ModuleNotFoundError: No module named 'requests'
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update>
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update>
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update> pip install requests
Collecting requests
  Downloading requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting charset_normalizer<4,>=2 (from requests)
```

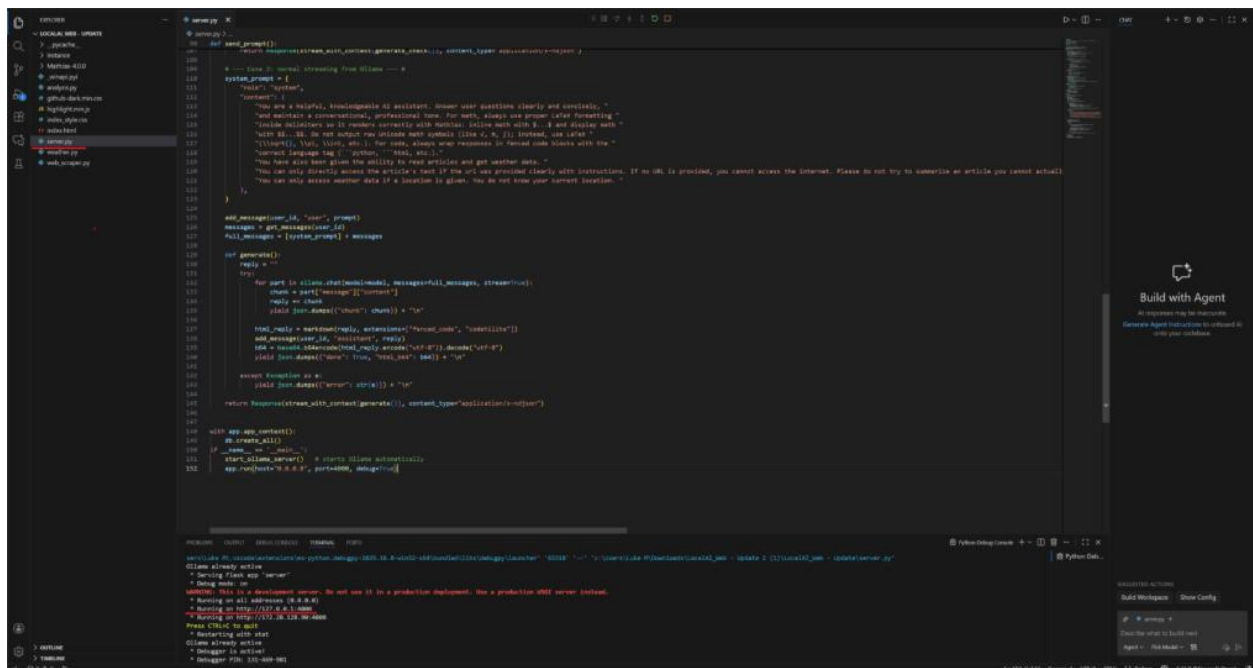
Type “*pip install request*” in the terminal. Then press control + f5 and if it works then you should be able to run the script.

```

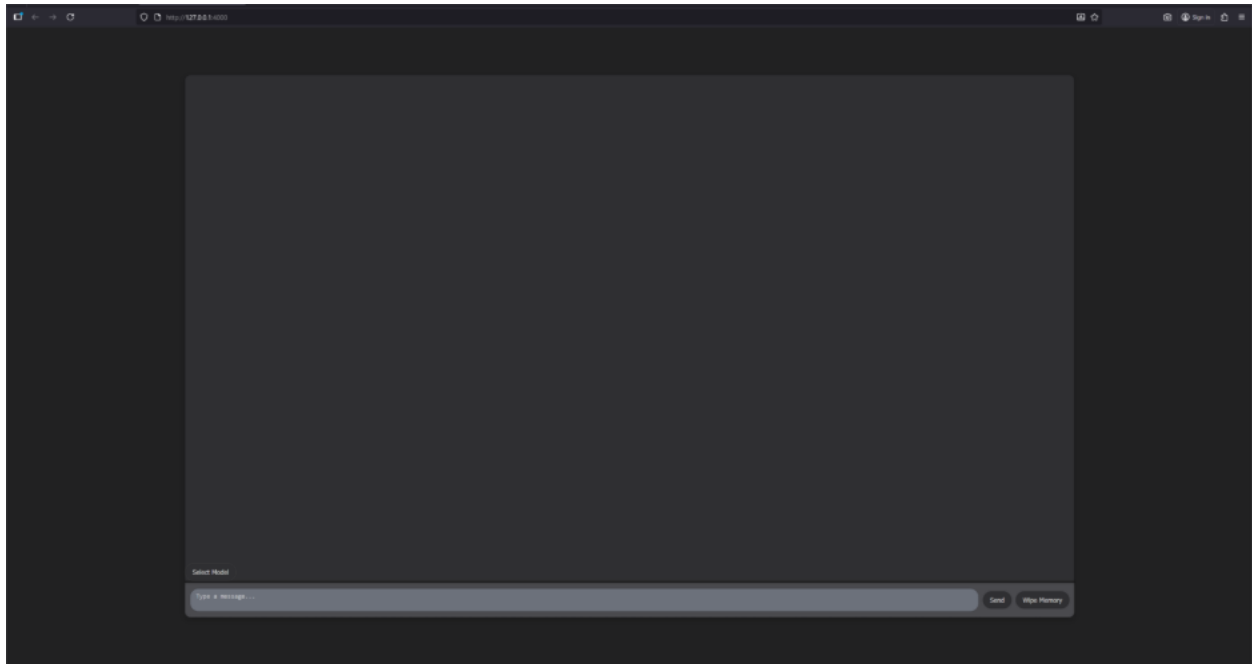
sers\Luke M\vscode\extensions\ms-python.debugpy-2025.16.0-win32-x64\bundled\libs\debugpy\launcher' '65299' '--' 'c:\Users\L
Traceback (most recent call last):
  File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\server.py", line 10, in <module>
    from analysis import analyze_prompt
  File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\analysis.py", line 5, in <module>
    from web_scraper import fetch_html
  File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\web_scraper.py", line 2, in <module>
    from bs4 import BeautifulSoup
ModuleNotFoundError: No module named 'bs4'
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update> pip install bs4
Collecting bs4
  Downloading bs4-0.0.2-py2.py3-none-any.whl.metadata (411 bytes)
Collecting beautifulsoup4 (from bs4)

```

When you pressed control +f5 check if there any module named bs4 and type “pip install bs4” if an error like underlined in the image above and the script should work.



Run the script with control + f5 and find the IP address underlined as highlighted in the image above.



Press control + click on the Ip address to open the AI on a web browser and an User Interface like the image above should be opened. Make sure you have the exact model of the llama application that is required, which is Llama3:8b. If you are not sure, then got to command prompt and type in `ollama run llama3:8b` to see if the specific llama is downloaded on your device.

Lab Summary

In this analysis/guide, we set up a locally hosted open source AI that allows any personnel such as teachers to keep the inputted data private without violating privacy laws like FERPA. The only notable requirements needed for this guide are basic python scripting knowledge or basic Linux Terminal Configuration knowledge if the personnel desires to change the default configurations provided in this guide/analysis. However, if the personnel does not need to change the default configurations then the technical skills mentioned in the previous sentence are not required.

Problems

One of the initial problems that I encountered was during the installation process of Ollama regarding the parameters for the model. Essentially, when installing Ollama you must specifically write the parameters after your `ollama run <AI model>` command otherwise the models will not properly function in the next steps. However, I did not know this piece of information causing me to have to reinstall all of the local models on my computer after realizing this.

On top of this, this entire installation process was done on a high use network causing just the download process by itself to take an extremely longer time than expected from the

low available bandwidth on the network. Also, I did not know that in Docker there is a command that must be entered to properly load the previous configurations causing me to have to remake the Docker container every time Docker was booted up. This was one of my first projects involving Linux, so at this point I did not have any knowledge of Linux terminal commands, which also delayed me when using Linux to install Ollama in the container. Later on, I even figured out there are even methods to make the AI models work without putting Ollama in the docker container, but in this guide we used Ollama in the container as it was the simplest method.

The last problem that I encountered was during the final process of the guide where Open WebUI and the Local AI were working but other people were not able to open the Web UI for the AI. Eventually, I found out that the reason was that when connecting to the same network as the AI, they would have their device's IP address automatically assigned as a random value causing issues, but this was fixed by going to the device's network settings and statically assigning the IP address to one that was compatible to the Web UI for the AI.

Conclusion

In this guide/analysis I explained and showed the step by step process for any personnel regardless of their familiarity with AI technology, how to set up an open source AI that is locally run on the personnel's network. This setup also includes how to create a website for the local AI that anyone on the same network and proper settings which anyone can access through Python or Docker. The main use for this guide is to have an Open-Sourced AI that is locally run allowing all data from the personnel to stay private instead of the normal company hosted AIs where all data is under the property of the company that created the AI which can be used and sold off.